

سومو

حالا با بانی

↑ sampling at the Nyquist Rate and signal reconstruction

$$1. \quad x_{e_1}(t) = \sum_{k \in \mathbb{Z}} (-1)^k x(kT_s) \delta(t - kT_s) = \sum_{k \in \mathbb{Z}} e^{j\pi k} x(kT_s) \delta(t - kT_s)$$

$$\Rightarrow F\{x_{e_1}(t)\} = X(f) * \frac{1}{T_s} \sum_{k \in \mathbb{Z}} \delta(f - \frac{k}{T_s}) = X(f) * H(f)$$

→ $H(f)$: scaled and shifted pulse train

→ $T_s < \frac{1}{2B}$ (Nyquist rate) → signal can be reconstructed. (Band pass filter from 0 to f_s)

$$2. \quad \text{sinc} \xrightarrow{f \cdot T} \text{rect}(t), \quad x(at) \rightarrow \frac{1}{a} x\left(\frac{t}{a}\right)$$

$$\Rightarrow 2B \text{sinc} \xrightarrow{f \cdot T} \text{rect}\left(\frac{t}{2B}\right)$$

$$x_p(t) = \sum x(kT_s) p(t - kT_s) = p * x$$

$$\rightarrow F\{x_p\} = X(f) P(f) = \text{sinc} \frac{f}{2B} \cdot f_s \sum_{k \in \mathbb{Z}} \text{rect} \frac{f - k f_s}{2B}$$

$$\rightarrow \boxed{\text{RPF}} \rightarrow y \Rightarrow y = f_s \text{rect} \frac{f}{2B} \text{sinc} \frac{f}{2B}$$

(just picking $n=0$) → maximum attenuation: $f=B$

$$\rightarrow y = f_s \text{sinc} \frac{1}{2}, \quad \text{sinc} \frac{1}{2} = \frac{\sin \frac{\pi}{2}}{\frac{\pi}{2}} = \frac{\sqrt{2}}{\pi}$$

2 Low pass signal sampled at the Nyquist Rate

$$x(t) = \sum_{k \in \mathbb{Z}} x(kT_s) \operatorname{sinc}\left(\frac{t - kT_s}{T_s}\right) \quad (\text{sinc interpolation})$$

$$x(nT_s) = \begin{cases} -1 & -\infty < n < 0 \\ 1 & 0 \leq n < \infty \\ 0 & \text{o.w} \end{cases}$$

$$t = 0/\omega = \frac{T_s}{f} \rightarrow x(0/\omega) = \sum_{n \in \mathbb{Z}} x[n] \operatorname{sinc}(0/\omega - n)$$

$$\operatorname{sinc} x = \operatorname{sinc}(-x)$$

$$\rightarrow x(0/\omega) = \operatorname{sinc} 0/\omega + \operatorname{sinc} 1/\omega + \operatorname{sinc} 2/\omega + \operatorname{sinc} 3/\omega$$

$$- \operatorname{sinc} 1/\omega - \operatorname{sinc} 2/\omega - \operatorname{sinc} 3/\omega - \operatorname{sinc} 4/\omega$$

$$\rightarrow x(0/\omega) = \operatorname{sinc} 0/\omega - \operatorname{sinc} 3/\omega$$

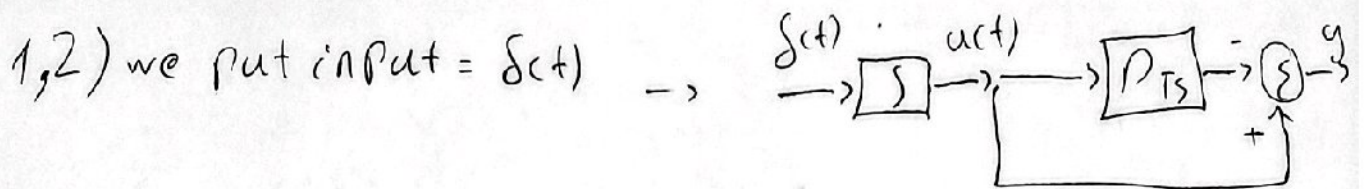
$$\rightarrow x(0/\omega) = \frac{2}{\pi} - \frac{2}{9\pi} = \frac{16}{9\pi} \approx 0.6$$

reconstructed $x(t)$ is a finite linear combination of sinc functions $\rightarrow$ E energy signal.

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} (x(t))^2 dt = \frac{1}{T_s} \sum_{n \in \mathbb{Z}} x(nT_s)^2 = 0.6$$

$$P = 0$$

3 Zero-order Hold circuit



$$\rightarrow y = u(t) - u(t - T_s) \rightarrow H(t) = \left(\frac{\delta(t)}{r} + \frac{1}{j\omega r} \right) (1 - e^{-jn\omega T_s})$$

$$= T_s \operatorname{sinc}\left(\frac{\omega T_s}{2}\right) e^{-j\omega T_s/2}$$

$$3) \bar{g}(t) = g(t) \sum_{n \in \mathbb{Z}} \delta(t - nT_s) = \sum_{n \in \mathbb{Z}} g(nT_s) \delta(t - nT_s)$$

$$\rightarrow y(t) = \sum_{n \in \mathbb{Z}} g(nT_s) (u(t - nT_s) - u(t - (n+1)T_s))$$

super position

4 PCM coding of compact Disc Audio

1. $B = 16 \text{ kHz}$ if $x(t) = 0$ for $|t| > B$

$\rightarrow$ reconstruction is possible if $f_s \geq 2B$

$\rightarrow f_{\text{Nyquist}} = 20 \text{ kHz}$

2. $r \geq L$, $L = 96000 \times 14 = 2^{14}$

$\rightarrow b = \log_2 L = \log_2 2^{14} = 14$

$\rightarrow 14 \text{ bits/sample}$

$$3. R_b = f_s \cdot b, f_s = f_N = 40 \text{ KHz}, b = 14$$

$$\rightarrow R_b = 560 \text{ Kbit/s}$$

$$4. R_b = f_s \cdot b = 5600 \times 14 = 78400 \text{ bits/s}$$

$$B_{\min} = \frac{R_b}{4} \rightarrow B_{\min} = 19600 \text{ KHz}$$

5 Quantization and SQNR in PCM

$$1. B = 6 \text{ KHz}, \text{Guard Band} = 1 \text{ KHz}, f = 12 \text{ K}$$

$$\frac{f_s}{4} = 6 + 1 = 7 \text{ KHz} \rightarrow f_s = 28 \text{ KHz}$$

$$\rightarrow R_b = 4(6+1) \log_2 16 = 96 \text{ Kbit/s}$$

$$2. \text{min Bandwidth} = \frac{96}{4} \text{ K} = 24 \text{ KHz}$$

$$3. P = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-1}^1 x^2 f(x) dx$$

$$= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2} \quad \Delta = \frac{1}{16}, \sigma_e^2 = \frac{\Delta^2}{12}$$

$$\rightarrow \text{SQNR}_{\text{dB}} = 10 \log_{10} 14.4 = 10 \log_{10} 14.4$$

$$\approx 11.5 \text{ dB}$$

$$4. \quad B = \frac{R_b}{2} \rightarrow R_{b, \max} = 180 \text{ K bit/s}$$

$$\rightarrow b_{\max} = \frac{R_{b, \max}}{f_s} = 10 \text{ bits/sample}$$

$$\rightarrow L_{\max} = \sqrt{10} = 10 \text{ KHz}$$

$$\Delta = \frac{\Sigma}{10^5} \rightarrow \sigma_{e, \max}^2 = \frac{(1/\sqrt{12})^2}{10^5} = \frac{1}{12 \times 10^5}$$

$$\text{SQNR}_{\max} = \frac{P_{\text{av}}}{\sigma_{e, \max}^2} = \frac{\frac{\Sigma}{2}}{\frac{1}{12 \times 10^5}} = 6 \times 10^6$$

$$\rightarrow \text{SQNR}_{\max, \text{dB}} = 10 \log_{10} 6 \times 10^6 \approx 60.8 \text{ dB}$$

6 FDM system Design

$$1. \quad D \triangleq \frac{\Delta f_i}{w_i} \quad (\text{deviation ratio is the same for all})$$

$$\rightarrow \Delta f_i = D w_i$$

$$\text{carlson rule: } B_i = 2(\Delta f_i + w_i)$$

$$\rightarrow B_i = 2(D w_i + w_i) = 2 \propto f_i (D+1)$$

$$\rightarrow B_i = \underbrace{2 \propto (D+1)}_{\text{some constant } A} f_i = A f_i \rightarrow B_i \propto f_i$$

$$\left(f_{i+1} - \frac{B_{i+1}}{r}\right) - \left(f_i + \frac{B_i}{r}\right) = B_g$$

$$B_i = A f_i, \quad B_{i+1} = A f_{i+1}$$

$$\rightarrow f_{i+1} \left(1 - \frac{A}{r}\right) = f_i \left(1 + \frac{A}{r}\right) + B_g$$

$$\rightarrow f_{i+1} = \frac{1+A/r}{1-A/r} f_i + B_g / (1-A/r) \quad \checkmark$$

$$\text{if } m_i(t) = 0 \text{ for } |t| > W_i$$

$$\rightarrow \text{sinc interpolation } m_i(t) = \sum_{n \in \mathbb{Z}} m_i(nT_i) \operatorname{sinc}\left(\frac{t-nT_i}{T_i}\right)$$

$$\text{where } T_i = \frac{1}{r W_i}$$

$$2. \quad f_1 = 2 \text{ KHz}, \quad B_1 = 100 \text{ Hz}, \quad B_g = 500 \text{ Hz}$$

$$B_i = f_i (B+1) \propto$$

$$f_{i+1} = 1/r f_i + \omega_{00} \text{ Hz}$$

$$\rightarrow f_r = f_1 / r + \omega_{00} = 2 \omega_{00} \text{ Hz}$$

$$f_r = 1/r f_r + \omega_{00} = \omega_{00} \text{ Hz}$$

$$f_g = 1/r f_r + \omega_{00} = \omega_{00} \text{ Hz} \dots$$

$$\text{if } f_{i+1} = r f_i + c, \quad r = 1/r, \quad c = \omega_{00}$$